Scoring systems for the prediction of choledocholithiasis: systematic review and meta-analysis (protocol)

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Short running title: The prediction model of choledocholithiasis

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1. Background

Choledocholithiasis is a major cause of hospitalization and the leading cause of symptomatic cholelithiasis and acute biliary pancreatitis [1]. Cholelithiasis and cholecystitis afflicts more than 20 million Americans annually at an estimated cost of $16.3 billion, resulting in 350,000 hospitalizations and 15,000 deaths [2]. The prevalence of choledocholithiasis is approximately 10% to 15% in patients with symptomatic cholelithiasis [3]. Up to one-third of patients with choledocholithiasis spontaneously excrete stones without intervention, but the majority of patients require endoscopic and/or surgical intervention [4]. Therefore, choledocholithiasis should be detected and managed as soon as possible [5].

The diagnosis of choledocholithiasis is challenging because the initial investigations including medical history, physical examination, laboratory tests, and transabdominal ultrasound (TUS) have low diagnostic accuracy [6, 7]. The next options of diagnostic examination for cholelithiasis is imaging such as magnetic resonance cholangiography (MRCP), endoscopic retrograde cholangiography (ERCP), endoscopic ultrasound (EUS), intraoperative cholangiography (IOC) [8]. MRCP is a non-invasive and accurate test compared to ERCP, but the main disadvantage of MRCP is that the identified common bile duct stones need to be removed by another intervention method [9]. EUS shows excellent diagnostic accuracy equivalent to MRCP, and EUS followed by ERCP is a cost-effective method because both EUS and ERCP can be performed in the same session [10]. However, ERCP is associated with adverse events in 6% to 15%, some life-threatening [11]. IOC is a gold standard technique for the diagnosis of intraoperative choledocholithiasis, but its disadvantages include radiation exposure, some failure rates, and increased surgery time [12].

Scoring systems for predicting choledocholithiasis has been attracting attention because the risk of choledocholithiasis can be evaluated, the next necessary tests and treatments can be decided, and unnecessary tests can be avoided. The American Society of Gastrointestinal Endoscopy (ASGE) published the algorithm that may be helpful for managing patients with suspected choledocholithiasis dependent on their risk stratification [13]. Some studies evaluating external validity have shown that the ASGE's algorithm has poor diagnostic accuracy [14–20]. After publication of the ASGE's algorithm, in the last years, at least four new scoring systems for the prediction of choledocholithiasis were published [21–24].

The present study will aim to assess the scoring systems for predicting choledocholithiasis.

2. Objectives

We will summarize the reported multivariable scoring systems for predicting choledocholithiasis (PART A), and assess the predictive performance of scoring systems for choledocholithiasis (PART B).

3. Methods

3.1. Protocol

To qualify the study, we will follow the prognostic model study definition proposed in the Critical Appraisal and Data Extraction for Systematic Reviews (CHARMS) [25]. We will publish this protocol in protocols.io (https://www.protocols.io/).

3.2 Criteria for considering studies for this review

3.2.1 Types of studies

We will include all studies on the development, the update, and/or the external validation of at least one multivariable model or scoring system for predicting choledocholithiasis. A multivariable model will be a mathematical equation that relates multiple predictors for participants with suspected choledocholithiasis to the probability of choledocholithiasis.

We will include studies reported as full text, those published as abstract only, and unpublished data. There is no restriction regarding publication language.

3.2.2 Types of participants

We will include studies in adults aged >18 years with suspected choledocholithiasis in any setting. We will exclude studies that look for risk factors for choledocholithiasis but have no score.

3.3 Outcomes

3.3.1 Primary outcomes

We will assess the quality of studies which develop or validate scoring systems for the prediction of choledocholithiasis.

We will assess the predictive performance of scoring systems for choledocholithiasis in model validation studies.

3.3.2 Secondary outcomes

We will review and appraise the validity and usefulness of the scoring systems for the prediction of choledocholithiasis.

We will assess reported performance of scoring systems for the prediction of choledocholithiasis.

3.4 Search methods for identification of studies

3.4.1 Electronic searches

We will search the following electronic databases:

1. Cochrane Central Register of Controlled Trials (CENTRAL); and
2. EMBASE (1988 to present).

We will also search the following clinical trial registers/trial result registers:

1. Clinical Trials.gov; and
2. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP).

We will check the reference lists of included studies and review articles for additional references. We will search the reference lists of guidelines for related to choledocholithiasis [13]. We will contact authors of identified trials and ask them to obtain additional data.

3.5 Data collection and analysis

3.5.1 Selection of studies

Two review authors (JW and TS) will independently screen titles and abstracts for inclusion. All of the potential studies we identify as a result of the search will be coded them as either 'retrieve' (eligible, potentially eligible, or unclear) or 'do not retrieve'. We will retrieve the full text of potentially eligible studies and two review authors (JW and TS) will independently screen the full text, and identify studies for inclusion, and identify and record reasons for exclusion of the ineligible studies. We will resolve any disagreement through discussion or, if required, we will consult a third author (TK and/or YK). We will identify and exclude duplicates and collate multiple reports of the same study, so that each study rather than each report is the unit of interest in the review. We will record the selection process in sufficient detail to complete a PRISMA flow diagram and 'Characteristics of excluded studies' table.

3.5.2 Data extraction and management

We will use a standardized data collection form for study characteristics and outcome data, which will be piloted on at least one study included in the review. Two review authors (JW and TS) will independently extract study characteristics from included studies.

We will extract the following data in accordance with the Critical Appraisal and Data Extraction for Systematic Reviews (CHARMS) [25]:source of data or study design (development and validation) (eg, prospective, case control, cohort, clinical consensus), study characteristics (development and validation) (eg, country, year, number of centers), participant characteristics (development and validation) (eg, number, age, sex, setting), predicted outcome (development and validation) (eg, choledocholithiasis diagnosis confirmed), model development (development) (eg, sample size, type of model (eg, logistic, neural network, machine learning techniques), handling of continuous variables (eg, complete-case analysis, imputation, or other methods), selection of variables, missing data, method of internal validation), model presentation (development) (eg, full regression model, simplified model, score chart), assessment of performance (development and validation) (eg, measures of discrimination (eg, c-statistic), measures of calibration (eg, expected/observed events), classification measures (eg, sensitivity, specificity, positive predictive value, negative predictive value)).

3.6 Assessment of risk of bias in included studies

Two review authors (JW and TS) will independently assess the risk of bias for each study included in the review, using the Prediction model Risk of Bias Assessment Tool (PROBAST) [26]. We will resolve any disagreement by discussion, or by involving a third review author (TK and/or YK). We will assess the risk of bias according to 23 signalling questions within four domains (participant selection, predictors, outcome and analysis). The models will be classed as low, high or unclear risk of bias.

3.7 Data synthesis

We will summarize the results using descriptive statistics and a narrative synthesis. We will not perform a quantitative synthesis of the scores or their predictive performance in model development studies. However, if we identify multiple studies that evaluate the same scoring system and report common performance measures in model validation studies, we will perform a random effects meta-analysis to calculate a summary estimate for models' performance and calibration. We will also consider for the meta-analysis those prediction models that will be internally validated through bootstrapping or cross validation and will be externally validated in only two independent datasets. We will follow a recently published framework for the meta-analysis of prediction models. If a measure of uncertainty (standard error or 95% confidence interval) is not available for mean C statistic, we will use a formula to approximate the standard error of mean C statistic based on number of events and number of participants. We will quantify between study heterogeneity by using the I2 and $\tau 2$ statistics.

We will perform the meta-analysis using RevMan 5.4.1 (The Cochrane Collaboration, Copenhagen, Denmark), Diagnostic Test Accuracy Meta-Analysis software, version 1.25 [27], or STATA software, version 16.0 (Stata Corporation, College Station, TX).

3.8 Assessment of heterogeneity

If we perform meta-analyses in model validation studies, to investigate the potential sources of heterogeneity, we will assess the effects of the following factors on the predictive performance of scoring systems for choledocholithiasis: setting (emergency department or patients admitted to hospital).

3.9 Sensitivity analysis

If we perform meta-analyses in model validation studies, we will present sensitivity analyses stratified by methodological quality as per the PROBAST. We will also perform the following sensitivity analyses to investigate the robustness of the results: excluding studies that their reference test is unknown.

4. Conflict of Interest

The authors declare no conflicts to interest.

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Supplement

Appendix 1: Search strategy

MEDLINE via Ovid

1. (((bile duct or biliary or CBD) adj5 (stone or stones or calculus or calculi)) or choledocholithiasis or cholelithiasis).ti,ab.

2. exp Gallstones/ or exp Choledocholithiasis/ or exp cholelithiasis/

3. 1 or 2

4. exp Clinical Decision Rules/

5. predict$ model$.ti,ab.

6. predict$ score.ti,ab.

7. predict$ value$.ti,ab.

8. clinical model$.ti,ab.

9. risk score$.ti,ab.

10. diagnostic score$.ti,ab.

11. model$ score$.ti,ab.

12. scoring system$.ti,ab.

13. algorithm.ti,ab.

14. validation.ti,ab.

15. stratification.ti,ab.

16. or/4-15

17. 3 and 16

Embase via Proquest

S1. ab(bile duct or biliary or CBD) OR ti(bile duct or biliary or CBD)

S2. ab(stone or stones or calculus or calculi) OR ti(stone or stones or calculus or calculi)

S3. S1 AND S2

S4. ab(choledocholithiasis or cholelithiasis) OR ti(choledocholithiasis or cholelithiasis)

S5. S3 OR S4

S6. EMB.EXACT.EXPLODE("common bile duct stone")

S7. EMB.EXACT.EXPLODE("bile duct stone")

S8. EMB.EXACT.EXPLODE("calcium stone (cholelithiasis)")

S9. S6 OR S7 OR S8

S10. S5 OR S9

S11. EMB.EXACT.EXPLODE("clinical decision rule")

S12. ab(prediction model) OR ti(prediction model) OR ab(prediction models) OR ti(prediction models) OR ab(predictive model) OR ti(predictive model) OR ab(predictive models) OR ti(predictive models)

S13. ab(prediction value) OR ti(prediction value) OR ab(prediction values) OR ti(prediction values) OR ab(predictive value) OR ti(predictive value) OR ab(predictive values) OR ti(predictive values)

S14. ab(clinical model) OR ti(clinical model) OR ab(clinical models) OR ti(clinical models)

S15. ab(risk score) OR ti(risk score) OR ab(risk scores) OR ti(risk scores)

S16. ab(diagnostic score) OR ti(diagnostic score) OR ab(diagnostic scores) OR ti(diagnostic scores)

S17. ab(model score) OR ti(model score) OR ab(model scores) OR ti(model scores)

S18. ab(scoring system) OR ti(scoring system) OR ab(scoring systems) OR ti(scoring systems)

S19. ab(algorithm) OR ti(algorithm)

S20. ab(validation) OR ti(validation)

S21. ab(stratification) OR ti(stratification)

S22. S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20

S23. S10 AND S22

ClinicalTrials.gov

Condition or disease: (bile duct) OR CBD OR choledocholithiasis OR cholelithiasis

other terms: predict OR rule OR score OR model OR value OR risk OR scoring systems OR algorithm OR validation OR stratification

ICTRP

((bile duct) OR CBD OR choledocholithiasis OR cholelithiasis)

AND

(predict OR rule OR score OR model OR value OR risk OR scoring systems OR algorithm OR validation OR stratification)